

Algebra

CHAPTER 4

Principle of Mathematical Induction

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Mathematical Induction

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Principle of Mathematical Induction

MATHEMATICAL INDUCTION

Mathematical induction is a technique for proving a statement, a theorem, or a formula, that is asserted about every natural number.

For example, formula to get the sum of first n natural numbers is

$$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

To verify this for $n = 1, n = 2, n = 3$, and so on, we can test the formula for any given number, say $n = 3$.

$$\text{L.H.S.} = 1 + 2 + 3 = 6$$

$$\text{and R.H.S.} = \frac{3(3+1)}{2} = 6$$

So, formula is true for $n = 3$.

Also, it can be verified that formula is true for $n = 4$.

But how do we verify this rule for every value of n ?

Here, the principle of mathematical induction comes into action.

It is the inference of the rule from the particular mathematical statements.

There are two steps to prove a mathematical statement using mathematical induction.

(i) The statement is true for 1.

(ii) If the statement is true for a natural number k , then it will also be true for its successor $(k + 1)$.

If both the steps are proved, then the statement will be true for every natural number n .

The hypothesis of step (ii) “The statement is true for k ” is called the induction assumption, or the induction hypothesis. This is what we assume when we prove a theorem by induction.

Let us understand the principle of mathematical induction in better way by proving the statement

$$P(n): 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2} \text{ for all natural numbers.}$$

For $n = 1$, we have

$$1 = \frac{1(1+1)}{2} \Rightarrow 1 = 1$$

So, the formula is true for $n = 1$ or $P(1)$ is true.

Now let us assume that the formula is true for $n = k$.

Then,

$$P(k) = 1 + 2 + 3 + \dots + k = \frac{k(k+1)}{2}$$

This is the induction assumption.

Assuming this, we must prove that the formula is true for its successor, $n = k + 1$.

So, we must show that

$$1 + 2 + 3 + \dots + (k + 1) = \frac{(k+1)(k+2)}{2}$$

We already have,

$$1 + 2 + 3 + \dots + k = \frac{k(k+1)}{2}$$

Adding $(k + 1)$ both sides, we get

$$1 + 2 + 3 + \dots + (k + 1) = \frac{k(k+1)}{2} + (k + 1)$$

$$\Rightarrow 1 + 2 + 3 + \dots + (k + 1) = \frac{(k+1)}{2}(k+2)$$

This shows that $P(k + 1)$ is true.

Hence, by mathematical induction, $P(n)$ is true all natural numbers.

Exercises

1. Using principle of mathematical induction, prove that for all $n \in \mathbb{N}$,

$$1 \times 3 + 3 \times 5 + 5 \times 7 + \dots + (2n-1) \times (2n+1) = \frac{n(4n^2 + 6n - 1)}{3}.$$

2. Using the principle of mathematical induction, prove that for all $n \in \mathbb{N}$,

$$\frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + \frac{1}{3 \cdot 4 \cdot 5} + \dots + \frac{1}{n(n+1)(n+2)} = \frac{n(n+3)}{4(n+1)(n+2)}$$

3. Using principle of mathematical induction, prove that for all $n \in \mathbb{N}$, $2^{3n} - 1$ is divisible by 7.
4. Using principle of mathematical induction, prove that for all $n \in \mathbb{N}$, $x^n - y^n$ is divisible by $x - y$, where x and y are any integers such that $x \neq y$.

5. Using principle of mathematical induction, prove that for

$$\text{all } n \in \mathbb{N}, n \geq 2, \sqrt{n} < \frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{n}}.$$

6. Using principle of mathematical induction, prove that for

$$\text{all } n \in \mathbb{N}, \frac{n^5}{5} + \frac{n^3}{3} + \frac{7n}{15} \text{ is a natural number.}$$

7. Using mathematical induction, prove that $7^{4^n} - 1$ is divisible by 2^{2n+3} for any natural number n .

8. Prove by mathematical induction that n^5 and n have the same unit digit for any natural number n .

9. A sequence $b_0, b_1, b_2, \dots$ is defined by letting $b_0 = 5$ and $b_k = 4 + b_{k-1}$, for all natural numbers k . Show that $b_n = 5 + 4n$, for all natural numbers n using mathematical induction.

Chapter 4

1. We have

$$P(n): 1 \times 3 + 3 \times 5 + 5 \times 7 + \dots + (2n-1) \times (2n+1)$$

$$= \frac{n(4n^2 + 6n - 1)}{3}$$

For $n = 1$,

$$\text{L.H.S.} = 1 \times 3 = 3;$$

$$\text{R.H.S.} = \frac{1 \cdot (4 + 6 - 1)}{3} = 3$$

Thus, $P(1)$ is holds.

Let $P(n)$ be true for some $n = k$.

$$1 \times 3 + 3 \times 5 + 5 \times 7 + \dots + (2k-1)(2k+1)$$

$$= \frac{k(4k^2 + 6k - 1)}{3} \quad \dots (1)$$

Adding $(2k+1)(2k+3)$ on both sides, we get

$$1 \times 3 + 3 \times 5 + 5 \times 7 + \dots + (2k-1)(2k+1) + (2k+1)(2k+3)$$

$$= \frac{k(4k^2 + 6k - 1)}{3} + (2k+1)(2k+3)$$

$$= \frac{k(4k^2 + 6k - 1) + 3(4k^2 + 8k + 3)}{3}$$

$$= \frac{4k^3 + 18k^2 + 23k + 9}{3}$$

$$= \frac{(k+1)(4k^2 + 14k + 9)}{3}$$

$$= \frac{(k+1)\{4(k+1)^2 + 6(k+1) - 1\}}{3}$$

Thus, $P(k+1)$ is true whenever $P(k)$ is true.

Hence, by the principle of mathematical induction, statement $P(n)$ is true for all natural numbers.

2. We have

$$P(n) = \frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + \frac{1}{3 \cdot 4 \cdot 5} + \dots + \frac{1}{n(n+1)(n+2)}$$

$$= \frac{n(n+3)}{4(n+1)(n+2)}$$

For $n = 1$,

$$\text{L.H.S.} = \frac{1}{1 \cdot 2 \cdot 3} = \frac{1}{6} \text{ and } \text{R.H.S.} = \frac{1 \cdot 4}{4 \cdot 2 \cdot 3} = \frac{1}{6}$$

Thus, $P(1)$ is true.

Let $P(n)$ be true for some $n = k$.

$$\text{Then } \frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + \frac{1}{3 \cdot 4 \cdot 5} + \dots + \frac{1}{k(k+1)(k+2)}$$

$$= \frac{k(k+3)}{4(k+1)(k+2)} \quad \dots (1)$$

Adding $\frac{1}{(k+1)(k+2)(k+3)}$ on both sides, we get

$$\frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + \frac{1}{3 \cdot 4 \cdot 5} + \dots + \frac{1}{k(k+1)(k+2)} + \frac{1}{(k+1)(k+2)(k+3)}$$

$$= \frac{k(k+3)}{4(k+1)(k+2)} + \frac{1}{(k+1)(k+2)(k+3)}$$

$$= \frac{1}{(k+1)(k+2)} \left\{ \frac{k(k+3)}{4} + \frac{1}{k+3} \right\}$$

$$= \frac{1}{(k+1)(k+2)} \left\{ \frac{k(k+3)^2 + 4}{4(k+3)} \right\}$$

$$= \frac{1}{(k+1)(k+2)} \left\{ \frac{k^3 + 6k^2 + 9k + 4}{4(k+3)} \right\}$$

$$= \frac{(k+1)^2(k+4)}{4(k+1)(k+2)(k+3)}$$

$$= \frac{(k+1)\{(k+1)+3\}}{4\{(k+1)+1\}\{(k+1)+2\}}$$

Thus, $P(k+1)$ is true whenever $P(k)$ is true.

Hence, by the principle of mathematical induction, statement $P(n)$ is true for all natural numbers.

3. Let $P(n): 2^{3n} - 1$ is divisible by 7.

For $n = 1$, $2^3 - 1 = 7$, which is divisible by 7.

Thus, $P(1)$ is true.

Let $P(n)$ be true for some $n = k$.

$$\text{Then } 2^{3k} - 1 = 7m, m \in \mathbb{N} \quad \dots (1)$$

$$\text{Now, } 2^{3(k+1)} - 1 = 8 \times 2^{3k} - 1$$

$$= 8(7m + 1) - 1 \quad [\text{Using (1)}]$$

$$= 56m + 7$$

$$= 7(8m + 1), \text{ which is divisible by 7.}$$

Thus, $P(k+1)$ is true whenever $P(k)$ is true.

So, by the principle of mathematical induction, $P(n)$ is true for all natural numbers.

4. Let $P(n): x^n - y^n$ is divisible by $x - y$, where x and y are any integers such that $x \neq y$, $n \in \mathbb{N}$.

For $n = 1$,

$$P(1) = x^1 - y^1 = x - y, \text{ which is divisible by } (x - y).$$

So, $P(1)$ is true.

Let $P(n)$ be true for some $n = k$.

Then $x^k - y^k$ is divisible by $(x - y)$.

$$\text{or } x^k - y^k = m(x - y), m \in \mathbb{N} \quad \dots (1)$$

$$\text{Now, } x^{k+1} - y^{k+1}$$

$$= x^k \times x - x^k \times y + x^k \times y - y^k \times y$$

$$= x^k(x - y) + y(x^k - y^k)$$

$$= x^k(x - y) + ym(x - y) \quad [\text{Using (i)}]$$

$$= (x - y)[x^k + ym], \text{ which is divisible by } (x - y)$$

Thus, $P(k+1)$ is true whenever $P(k)$ is true.

So, by the principle of mathematical induction, $P(n)$ is true for any natural number n .

5. Let $P(n) : \sqrt{n} < \frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{n}}$, for all natural numbers, $n \geq 2$.

For $n = 2$, $\sqrt{2} < \frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}}$, which is true.

So, $P(2)$ is true.

Let $P(n)$ be true for some $n = k$.

$$\text{Then } \sqrt{k} < \frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{k}} \quad \dots(1)$$

Adding $\frac{1}{\sqrt{k+1}}$ on both sides, we get

$$\sqrt{k} + \frac{1}{\sqrt{k+1}} < \frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{k}} + \frac{1}{\sqrt{k+1}} \quad \dots(2)$$

$$\text{Now, } \sqrt{k+1} < \sqrt{k} + \frac{1}{\sqrt{k+1}} \quad \dots(3)$$

$$\text{or } k+1 < \sqrt{k} \sqrt{k+1} + 1$$

$$\text{or } k < \sqrt{k^2 + k}, \text{ which is true.}$$

So, from (2) and (3),

$$\sqrt{k+1} < \frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{k}} + \frac{1}{\sqrt{k+1}}$$

Thus, $P(k+1)$ is true whenever $P(k)$ is true.

So, by the principle of mathematical induction, $P(n)$ is true for any natural number n .

6. Let $P(n) : \frac{n^5}{5} + \frac{n^3}{3} + \frac{7n}{15}$ is a natural number, for all $n \in N$.

$$P(1) : \frac{1^5}{5} + \frac{1^3}{3} + \frac{7(1)}{15} = \frac{3+5+7}{15} = \frac{15}{15} = 1, \text{ which is a natural number.}$$

Hence, $P(1)$ is true.

Let $P(n)$ be true for some $n = k$.

$$\text{Then } \frac{k^5}{5} + \frac{k^3}{3} + \frac{7k}{15} \text{ is natural number.}$$

$$\begin{aligned} \text{Now, } \frac{(k+1)^5}{5} + \frac{(k+1)^3}{3} + \frac{7(k+1)}{15} \\ = \frac{k^5 + 5k^4 + 10k^3 + 10k^2 + 5k + 1}{5} \\ + \frac{k^3 + 1 + 3k^2 + 3k}{3} + \frac{7k + 7}{15} \end{aligned}$$

$$\begin{aligned} &= \frac{k^5}{5} + \frac{k^3}{3} + \frac{7k}{15} + k^4 + 2k^3 + 3k^2 + 2k + 1 \\ &= P(k) + k^4 + 2k^3 + 2k^2 + 2k + 1 \\ &= \text{Natural number} \end{aligned}$$

Thus, $P(k+1)$ is true whenever $P(k)$ is true.

So, by the principle of mathematical induction, $P(n)$ is true for any natural number n .

7. Let $P(n) = 7^{4^n} - 1$ be divisible by 2^{2n+3} .

$$\begin{aligned} P(1) &= 7^4 - 1 = (7^2 - 1)(7^2 + 1) \\ &= 48 \times 50 = 32 \times 75 \\ &= 2^5 \times 75, \end{aligned}$$

which is divisible by $2^{2 \times 1 + 3}$

Let us assume that the result is true for $n = k$.

i.e., $7^{4^k} - 1$ is divisible by 2^{2k+3} but not by 2^{2k+4} .

$$\Rightarrow 7^{4^k} - 1 = 2^{2k+3} m; \text{ where } m \text{ is some odd natural number}$$

$$\begin{aligned} \text{Now, } 7^{4^{k+1}} - 1 &= (7^{4^k})^4 - 1 \\ &= ((2^{2k+3} m + 1)^2 + 1)((2^{2k+3} m + 1)^2 - 1) \\ &= ((2^{2k+3} m + 1)^2 + 1)(2^{2k+3} m + 2)(2^{2k+3} m) \\ &= (2^{4k+6} m^2 + 2^{2k+4} m + 2)(2^{2k+3} m + 2)(2^{2k+3} m) \\ &= 2^{2k+5}(2^{4k+5} m^2 + 2^{2k+3} m + 1)(2^{2k+2} m + 1)(m); \end{aligned}$$

which is divisible by 2^{2k+5} .

Thus, $P(k+1)$ is true whenever $P(k)$ is true.

So, by the principle of mathematical induction, $P(n)$ is true for any natural number n .

8. We have to prove that $n^5 - n$ is divisible by 10.

$$\text{For } n = 1, 1^5 - 1 = 0 \text{ is divisible by 10.}$$

$$\text{Also, } n = 2, 2^5 - 2 = 30 \text{ is divisible by 10.}$$

Thus, $P(1)$ and $P(2)$ are true.

Let $P(n)$ be true for some $n = k$.

$$k^5 - k = 10m \quad \dots(1)$$

$$\begin{aligned} \text{Now, } (k+1)^5 - (k+1) &= k^5 + 5k^4 + 10k^3 + 10k^2 + 5k + 1 - k - 1 \\ &= (k^5 - k) + 5k(k^3 + 1) + 10k^3 + 10k^2 \\ &= 10m + 10k^3 + 10k^2 + 5k(k^3 + 1) \quad [\text{Using (1)}] \end{aligned}$$

Clearly, $k(k^3 + 1)$ is even for $k \in N$.

Thus, $P(k+1)$ is true whenever $P(k)$ is true.

So, by the principle of mathematical induction, $P(n)$ is true for any natural number n .

9. We have a sequence $b_0, b_1, b_2, \dots$ is defined by letting $b_0 = 5$ and $b_k = 4 + b_{k-1}$, for all natural numbers k .

$$\text{Let } P(n) : b_n = 5 + 4n, \text{ for all natural numbers } n$$

For $n = 1$,

$$b_1 = 5 + 4 \times 1 = 9$$

$$\text{Also } b_0 = 5$$

$$\therefore b_1 = 4 + b_0 = 4 + 5 = 9$$

Thus, $P(1)$ is true.

Now, assume that $P(n)$ is true for $n = k$.

$$\therefore P(k) : b_k = 5 + 4k \quad \dots(1)$$

Now, to prove $P(k+1)$ is true, we have to show that

$$\begin{aligned} P(k+1) : b_{k+1} &= 5 + 4(k+1) \\ b_{k+1} &= 4 + b_{k+1-1} \quad (\text{as } b_k = 4 + b_{k-1}) \\ &= 4 + b_k \\ &= 4 + 5 + 4k = 5 + 4(k+1) \end{aligned}$$

Hence, $P(k+1)$ is true whenever $P(k)$ is true.

So, by the principle of mathematical induction $P(n)$ is true for any natural number n .